Viva Questions for Practicals

SL3 Lab Manual.pdf
PDF

Gimme Viva Questions and answers for each Practicals

Software Laboratory III – Viva Questions & Answers

Based on your lab manual SL3 Lab Manual

Practical 1: Data Wrangling I

1. What is Data Wrangling?

Data Wrangling is the process of cleaning, transforming, and organizing raw data into a usable format for analysis and machine learning.

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2. Why is Data Wrangling important?

It improves data quality, removes inconsistencies, handles missing values, and prepares data for analysis and model building.

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3. What are the steps of Data Wrangling?

1. Discovering
2. Structuring
3. Cleaning
4. Enriching
5. Validating
6. Publishing

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4. What is Data Preprocessing?

Data preprocessing is the technique of converting raw data into a clean and understandable format before analysis.

5. What is normalization?

Normalization scales numeric values into a common range so that no feature dominates others.

6. Explain Min-Max Normalization.

It converts values between 0 and 1 using:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

7. What is Z-score normalization?

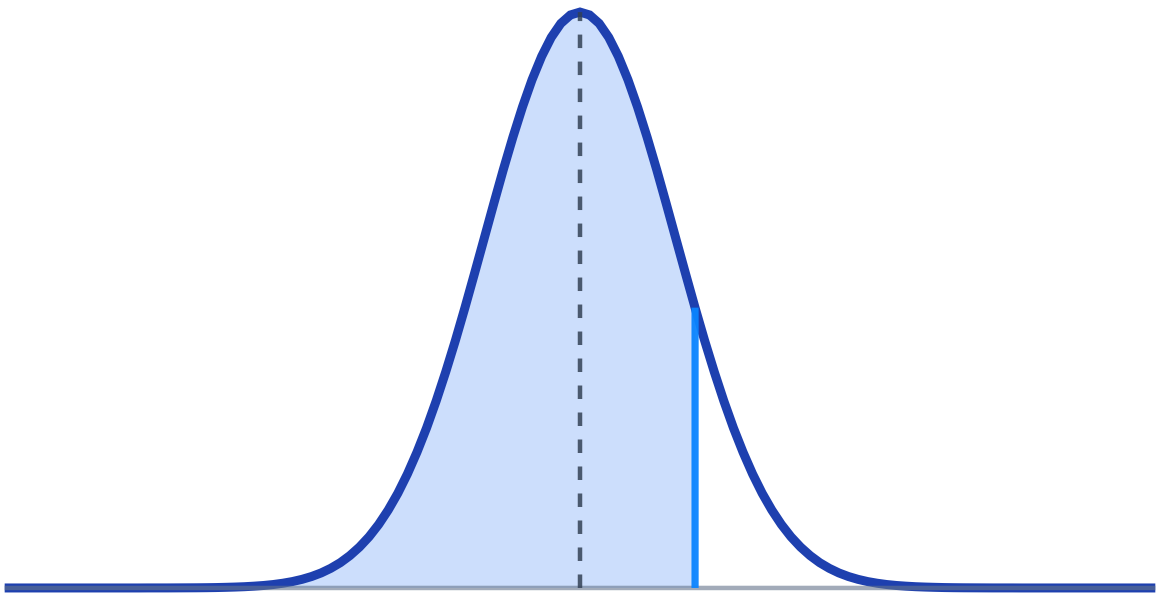
It standardizes data using mean and standard deviation.

$$Z = \frac{x - \mu}{\sigma}$$

x 1.2 μ 0.0 σ 1.0

$z = \frac{x-\mu}{\sigma} \approx 1.2$

$\Phi(z) \approx 88.5\%$



8. Difference between `info()` and `describe()`?

info()	describe()
Gives datatype and null values	Gives statistical summary
Used for structure analysis	Used for numerical analysis

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9. What is Pandas?

Pandas is a Python library used for data manipulation and analysis.

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10. What is a DataFrame?

A DataFrame is a 2-dimensional tabular data structure in Pandas.

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11. How do you check missing values?

Using:

```python
df.isnull()
```

12. How do you check dimensions of dataset?

```python
df.shape
```

13. What is One Hot Encoding?

It converts categorical variables into binary columns.

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14. What is Label Encoding?

It converts categorical labels into numeric form.

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15. Difference between categorical and numerical data?

Categorical	Numerical
Text labels	Numbers
Example: Gender	Example: Age

Practical 2: Data Wrangling II

1. What are missing values?

Missing values are empty or unavailable entries in a dataset.

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2. How can you handle missing values?

1. Remove rows
2. Replace using mean
3. Replace using median
4. Replace using mode

---

3. What is `fillna()`?

It fills missing values with specified data.

---

4. What is `dropna()`?

It removes rows or columns containing null values.

---

5. What are outliers?

Outliers are abnormal data points that differ significantly from others.

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6. How can outliers be detected?

1. Box Plot
2. Scatter Plot
3. Z-score

7. What is IQR?

Inter Quartile Range measures spread of middle 50% data.

$IQR = Q3 - Q1$

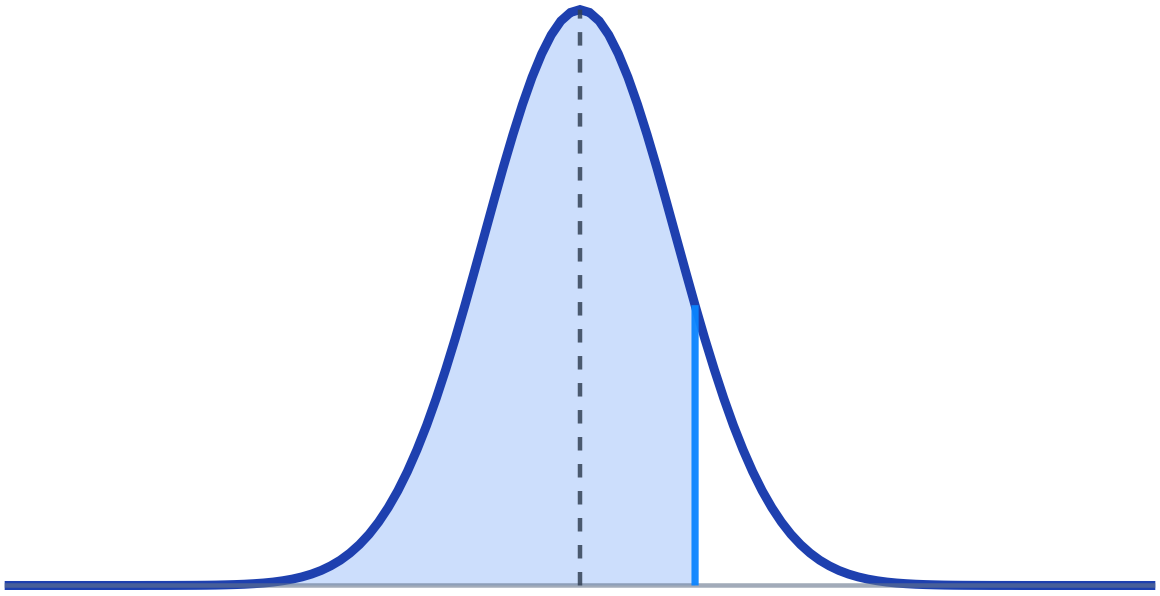
8. Formula for Z-score?

$Z = \frac{x-\mu}{\sigma}$

x 1.2 μ 0.0 σ 1.0

$z = \frac{x-\mu}{\sigma} \approx 1.2$

$\Phi(z) \approx 88.5\%$



9. What is data transformation?

It converts data into suitable format for analysis.

10. Difference between normalization and standardization?

Normalization	Standardization
Scales between 0 and 1	Mean = 0 and SD = 1
Used when no Gaussian distribution	Used for normal distribution

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11. What is skewness?

Skewness measures asymmetry in data distribution.

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12. What is positively skewed data?

Data with long tail on the right side.

---

13. What is negatively skewed data?

Data with long tail on the left side.

---

14. What is RobustScaler?

It scales data using median and IQR, useful for outliers.

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15. What is StandardScaler?

It standardizes data using mean and standard deviation.

Practical 3: Descriptive Statistics

1. What is Statistics?

Statistics is the science of collecting, organizing, analyzing, and interpreting data.

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2. Types of Statistics?

1. Descriptive Statistics
2. Inferential Statistics

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3. What is Mean?

$\bar{x} = \frac{\sum x}{n}$

Average of all observations.

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4. What is Median?

Middle value after sorting data.

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5. What is Mode?

Most frequently occurring value.

---

6. What is Standard Deviation?

Measure of spread of data from mean.

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7. What is Variance?

$\sigma^2 = \frac{\sum (x-\mu)^2}{N}$

---

8. What is `describe()` in pandas?

It gives statistical summary like mean, std, min, max, percentiles.

---

9. What is `groupby()`?

It groups data based on categories and performs aggregation.

---

10. What is Iris Dataset?

A famous ML dataset containing flower measurements of Iris species.

11. Which libraries are used?

- Pandas
- NumPy
- Matplotlib

12. Difference between population and sample?

Population	Sample
Entire data	Small subset

Practical 4: Linear Regression

1. What is Machine Learning?

Machine learning allows systems to learn from data without explicit programming.

2. Types of Machine Learning?

1. Supervised
2. Unsupervised
3. Reinforcement Learning

3. What is Regression?

Regression predicts continuous values.

4. What is Linear Regression?

Linear Regression finds best-fit straight line between variables.

5. Linear Regression equation:

$y = b_0 + b_1 x$

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6. What is dependent variable?

Variable being predicted.

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7. What is independent variable?

Input variable used for prediction.

---

8. What is slope?

Rate of change in dependent variable.

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9. What is intercept?

Value of y when x = 0.

---

10. What is MSE?

$MSE = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$

---

11. Difference between loss function and cost function?

Loss Function	Cost Function
Error for one sample	Average error for all samples

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12. Which library is used for Linear Regression?

`sklearn.linear_model`

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13. Which type of ML uses Linear Regression?

Supervised Learning.

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14. What is best fit line?

Line having minimum error.

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15. What is Boston Housing Dataset?

Dataset containing housing price information of Boston city.

Practical 5: Logistic Regression

1. What is Logistic Regression?

A classification algorithm used for binary outcomes.

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2. What is binary classification?

Classification with only two classes.

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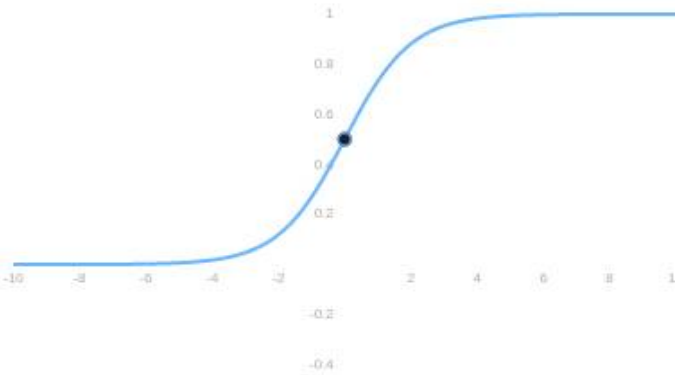
3. What is Sigmoid Function?

$f(x) = \frac{1}{1+e^{-x}}$



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4. Why sigmoid is used?

It converts values into probabilities between 0 and 1.

5. Difference between Linear and Logistic Regression?

Linear Regression	Logistic Regression
Predicts continuous output	Predicts categorical output

6. What is Confusion Matrix?

Table showing prediction performance.

7. What is True Positive?

Correctly predicted positive values.

8. What is False Positive?

Incorrectly predicted positive values.

9. What is True Negative?

Correctly predicted negative values.

10. What is False Negative?

Incorrectly predicted negative values.

11. Accuracy Formula

$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$

12. Precision Formula

$Precision = \frac{TP}{TP+FP}$

13. Recall Formula

$Recall = \frac{TP}{TP+FN}$

14. What is F1-score?

$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$

15. Applications of Logistic Regression

- Spam detection
- Cancer prediction
- Customer churn prediction

Practical 6: Naive Bayes

1. What is Naive Bayes?

A probabilistic classification algorithm based on Bayes theorem.

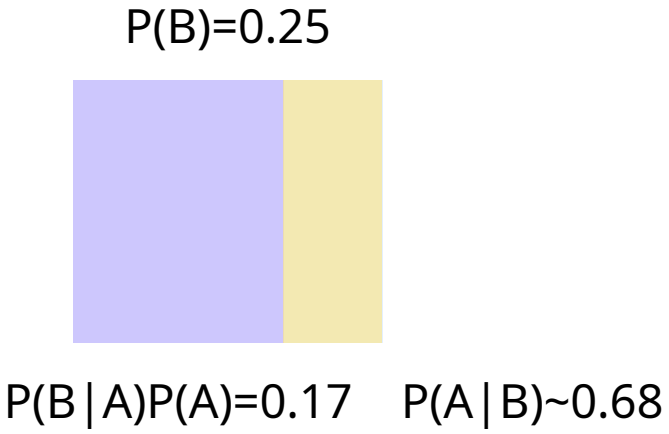
2. Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$P(A)$ 0.20 $P(B \mid A)$ 0.85 $P(B \mid \neg A)$ 0.10

$P(A \mid B) = \frac{P(B|A)P(A)}{P(B)} \approx 0.68, \; P(B) \approx 0.25$

Posterior = useful evidence / total evidence



3. Why is it called Naive?

Because it assumes all features are independent.

4. Applications of Naive Bayes

- Spam filtering
- Sentiment analysis
- Text classification

5. Advantages of Naive Bayes

- Fast
- Simple

- Works well with text data

6. What is prior probability?

Probability before observing evidence.

7. What is posterior probability?

Probability after observing evidence.

8. Which dataset is used?

Iris Dataset.

9. Which library is used?

`sklearn.naive_bayes`

10. Difference between Logistic Regression and Naive Bayes?

Logistic Regression	Naive Bayes
Discriminative model	Generative model
Uses sigmoid	Uses probability theorem

Practical 7: Text Analytics

1. What is NLP?

Natural Language Processing helps computers understand human language.

2. What is Tokenization?

Breaking text into words or tokens.

3. What are stop words?

Common words removed during preprocessing.

4. What is stemming?

Reducing words to root form.

5. What is lemmatization?

Converting words into meaningful root form.

6. Difference between stemming and lemmatization?

Stemming	Lemmatization
Faster	More accurate
Cuts words	Uses vocabulary

7. What is POS tagging?

Assigning grammatical tags to words.

8. What is TF-IDF?

Technique to measure word importance.

9. TF Formula

$$TF = \frac{\text{Number of times term appears}}{\text{Total terms}}$$

10. IDF Formula

$$IDF = \log\left(\frac{N}{df}\right)$$

Practical 8–10: Data Visualization

1. What is Data Visualization?

Representation of data using graphs and charts.

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2. Which library is used?

- Matplotlib
- Seaborn

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3. What is Histogram?

Graph showing frequency distribution.

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4. What is Box Plot?

Graph showing quartiles and outliers.

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5. What is Scatter Plot?

Shows relationship between two variables.

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6. What are outliers in boxplot?

Values outside whiskers.

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7. Why is visualization important?

Helps understand patterns and trends easily.

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8. What is Titanic Dataset?

Dataset containing passenger details of Titanic ship.

---

9. What is Iris Dataset used for?

Classification and visualization tasks.

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10. Difference between histogram and bar chart?

Histogram	Bar Chart
Continuous data	Categorical data

Practical 11–13: Hadoop & MapReduce

1. What is Big Data?

Extremely large and complex data.

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2. What is Hadoop?

Framework for distributed storage and processing.

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3. What is HDFS?

Hadoop Distributed File System.

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4. What is MapReduce?

Programming model for processing large datasets.

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5. Explain Mapper.

Processes input data and generates key-value pairs.

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6. Explain Reducer.

Aggregates mapper output.

7. What is WordCount?

Counts occurrences of words in text.

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8. What is distributed computing?

Using multiple systems together for computation.

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9. Advantages of Hadoop

- Scalable
- Fault tolerant
- Cost effective

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10. What is NameNode?

Master node managing metadata in Hadoop.

These viva questions are prepared from your SL-III Lab Manual

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